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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Sept – 2024 Examinations

Department of Artificial Intelligence and Machine Learning

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

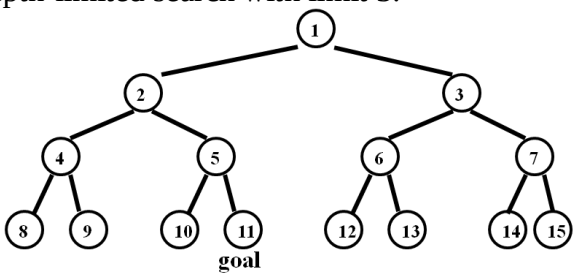
(2022 SCHEME)

Model Question Paper

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer ALL full questions from part B.

S. No	PART A	Marks
1.1	Identify the various capabilities that a computer has to possess to pass Turing test	01
1.2	Define Rational Agent (RA) and discuss the usage of RA in Healthcare applications	02
1.3	For each of the following agents, develop a PEAS description of the task environment: a. Robot soccer player b. Internet book-shopping agent	02
1.4	For the real time "AIRLINE" travel problem, identify the five components involved in problem formulation in-terms of problem-solving agents.	01
1.5	Consider the following state space shown in Fig.1.5. Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search and depth-limited search with limit 3.  Fig.1.5: State space	02
1.6	Analyze and identify issues in following machine learning scenarios and propose solution(s) for them: a. A student dataset has an employee age value as 230. b. An object recognition model training set error is 95% and test set error is 15%.	02
1.7	The average squared difference between classifier predicted output and actual output is called as_____	01
1.8	In a decision tree algorithm, how can you reduce the level of entropy from the root node to the leaf node?	01

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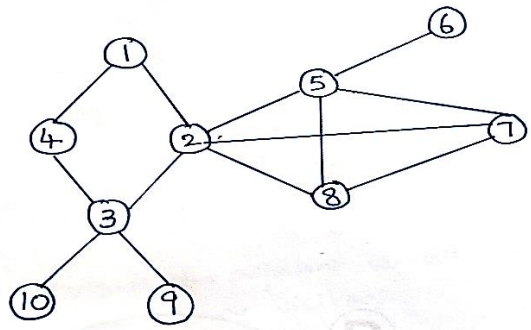
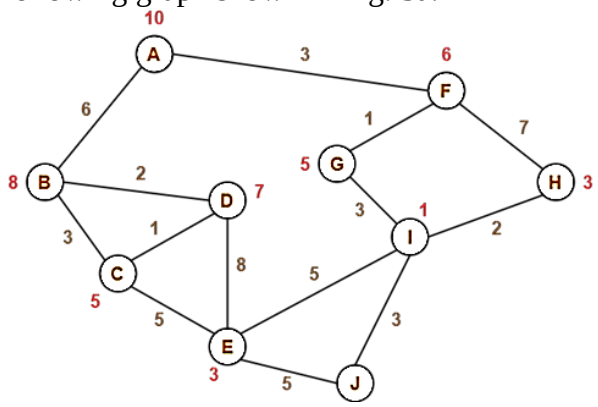
1.9	In the decision tree, the measure of the degree of probability of a particular variable being wrongly classified when it is randomly chosen is called _____.	01
1.10	While predicting the spam email using a classification model following data are recorded. i. Correct prediction – 0 spam, 75 non-spam ii. Incorrect prediction- 30 spam, 700 non-spam Calculate Accuracy and sensitivity.	02
1.11	What are some of the hyperparameters of the random forest regressor that help prevent overfitting?	02
1.12	There are a number of choices for the proximity function, centroid, and objective function that can be used in the basic K-means algorithm and that are guaranteed to converge. What are they?	02
1.13	How do you handle missing or corrupted data in a dataset?	01
	PART B	
2a.	Explain how AI works in the following areas: i. Google Search Engine ii. Voice Assistants	04
2b.	Justify with reasons how utility-based agents is different from model-based reflex agents with appropriate diagrams.	06
2c.	Apply the Depth first search algorithm for the following graph shown in Fig. 2c.  Fig. 2c a. Display the output visiting each node, and b. Develop the DFS spanning tree.	06
3a.	Consider the following graph shown in Fig. 3a.  Fig. 3a Find the most cost-effective path to reach from start state A to final state J using A-star Algorithm.	06

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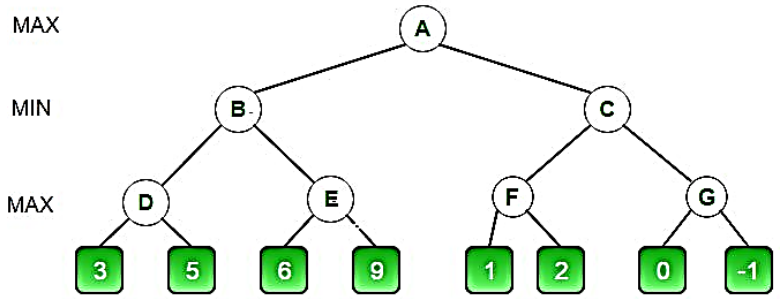
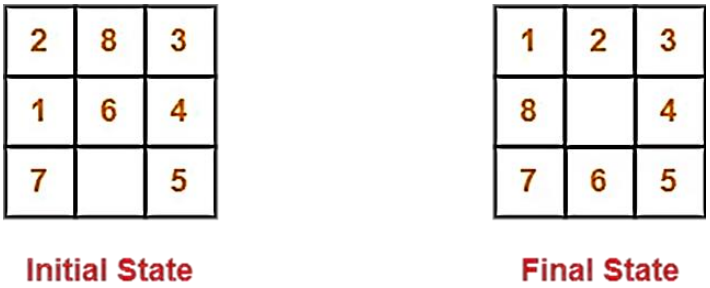
3b.	For the following two-ply game tree shown in Fig. 3b, the terminal nodes show the utility values computed by the utility function. Use the Min-Max algorithm to compute the utility values for other nodes in the given game tree.  Fig. 3b	10
	OR	
4a.	Given an initial state of 8-puzzle problem and final state to be reached-  Fig. 4a Find the most cost-effective path to reach the final state from initial state using A-star Algorithm for the Fig. 4a.	10
4b.	With neat sketches discuss the following classical search algorithms: i. Hill climbing ii. Simulated Annealing	06
5a.	Identify the key differences between predictive and descriptive machine learning modeling techniques in terms of approach, applications, algorithms used, evaluation measures.	04
5b.	Illustrate the bias-variance trade-off issue in machine learning. Identify the techniques used to avoid high bias and high variance scenarios.	06
5c.	Analyze and identify issues in the following machine learning scenarios and propose solutions (s) for them i. A student dataset has an employee age value of 230. ii. An object recognition model training set error is 95%, and the test set error is 15%. iii. Malignant tumour classification model training set error is 25%, and test set error is 23%.	06
	OR	
6a.	For instance, the following table informs about decision making factors to play tennis at outside for previous 14 days:	10

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Day	Outlook	Temp	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

i. Calculate the IG (information gain) for each attribute (feature)
ii. Choose a feature for each node to split
iii. Build Final Tree

6b. The following data are recorded while predicting the spam email using a classification model. (06)

i. Correct prediction – 0 spam, 75 non-spam
ii. Incorrect prediction- 30 spam, 700 non-spam

Calculate the Accuracy, sensitivity, precision, and F-measure of the model. Identify the problem with the above classification model.

7a. With relevant mathematical equations discuss Naïve Bayes Classification. Consider the following data in table 7a. How would a Naive Bayes classifier used to classify the sample X (Cow, Medium, Black)? Write step-by-step approach to solution. (10)

Table 7a. Dataset

	Animals	Size of Animal	Body Color	Can we Pet them
0	Dog	Medium	Black	Yes
1	Dog	Big	White	No
2	Rat	Small	White	Yes
3	Cow	Big	White	Yes
4	Cow	Small	Brown	No
5	Cow	Big	Black	Yes
6	Rat	Big	Brown	No
7	Dog	Small	Brown	Yes
8	Dog	Medium	Brown	Yes
9	Cow	Medium	White	No
10	Dog	Small	Black	Yes
11	Rat	Medium	Black	No
12	Rat	Small	Brown	No
13	Cow	Big	White	Yes

7b. For the given dataset shown in Table 7b, apply Logistic Regression. (06)

Table 7b.

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Hours Study	Pass (1)/Fail (0)
29	0
15	0
33	1
28	1
39	1

i. Assume the model suggested by the optimizer for odds of posing the course is: log(odds) = -64+2*hours.
ii. Given the class, calculate the probability of pass percentage for the student who studies for 33 hours.
iii. Compute the minimum number of hours of preparation student should do to pass the course with the probability of > 95%.

OR

8a. Reflect on working of KNN algorithm. Consider the data provided in Table 8a, using KNN algorithm with value of k = 3, find the weight for data: Height= 5.5 and Age= 38. Write step-by-step approach to solution. (10)

Table 8a dataset

Height	Age	Weight
5	45	59
5.11	46	77
5.6	55	61
5.7	56	65
5.4	28	50
5.3	35	53
5.0	46	59
4.11	26	67
5.3	57	57
5.5	43	55

8b. With neat sketches, explain boosting and bagging ensemble learning techniques. (06)

9a. Distinguish between the following: (09)
i. Hierarchical versus Partitional
ii. Exclusive versus Overlapping versus Fuzzy
iii. Complete versus Partial

9b. Illustrate and explain Bisecting K-means algorithm. (07)

OR

10 a. Given fifteen points in the Cartesian coordinate system and the requirement to form 3 clusters (K=3), if 3 centroids are randomly selected from the data—specifically the points A2 (2, 6), A7 (5, 10), and A15 (6, 11) respectively, how would the process of clustering proceed over 3 iterations? (10)

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		Point	Coordinates	Point	Coordinates	
		A1	(2,10)	A8	(4,9)	
		A2	(2,6)	A9	(10,12)	
		A3	(11,11)	A10	(7,5)	
		A4	(6,9)	A11	(9,11)	
		A5	(6,4)	A12	(4,6)	
		A6	(1,2)	A13	(3,10)	
		A7	(5,10)	A14	(3,8)	
				A15	(6,11)	
10 b.	How can cohesion and separation metrics be used to evaluate the quality of clusters in an unsupervised learning context?					06

Signature of Scrutinizer: Signature of Chairman
Name: Name: